

Initial Project Planning Template

Date	04 July 2026
Team ID	SOLO-CCFD-2026
Project Name	Credit Card Fraud Detection using Machine Learning
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation — 4 sprints of ~1 week each, solo development (team members did not respond, so all sprints executed individually).

Sprint	Functional Req (Epic)	User Story #	User Story / Task	Points	Priority
Sprint-1	Data Collection & EDA	USN-1	As a developer, I can download and load the ULB credit card transaction dataset for analysis	2	High
Sprint-1	Data Collection & EDA	USN-2	As a developer, I can explore class imbalance, feature distributions, and correlations via EDA	2	High
Sprint-1	Data Preprocessing	USN-3	As a developer, I can scale the Time and Amount features so they're comparable to the PCA components	1	High
Sprint-1	Data Preprocessing	USN-4	As a developer, I can apply SMOTE to balance the extremely skewed fraud/genuine classes	2	High
Sprint-2	Model Building	USN-5	As a developer, I can train a Logistic Regression baseline classifier	1	High
Sprint-2	Model Building	USN-6	As a developer, I can train a Decision Tree classifier	1	Medium
Sprint-2	Model Building	USN-7	As a developer, I can train a Random Forest classifier	2	High
Sprint-2	Model Building	USN-8	As a developer, I can train an XGBoost classifier	2	High
Sprint-3	Model Evaluation	USN-9	As a developer, I can compare all 4 models using precision, recall, F1, and AUPRC to pick the best one	2	High
Sprint-3	Model Evaluation	USN-10	As a developer, I can save the best-performing model and scaler to disk for deployment	1	High
Sprint-3	Web Application	USN-11	As an analyst, I can enter a transaction's features into a web form and get an instant fraud/genuine prediction	2	High
Sprint-3	Web Application	USN-12	As a compliance officer, I can upload a CSV of transactions and get batch predictions	2	Medium
Sprint-4	Deployment	USN-13	As a developer, I can deploy the trained model to IBM Watson Machine Learning	2	High
Sprint-4	Deployment	USN-14	As a user, I can access the Flask app publicly via a hosted/tunnelled URL for demonstration	1	Medium
Sprint-4	Testing & Documentation	USN-15	As a developer, I can test the deployed app's performance and document the full solution	2	High

Sprint Summary: Sprint 1 — Data collection, EDA & preprocessing. Sprint 2 — Model building (4 algorithms). Sprint 3 — Model evaluation, selection, and Flask web app. Sprint 4 — IBM Watson ML deployment, public access, testing & documentation.